

Physics 2

Comprehensive Review Package

Professor-first master review and reusable knowledge base

Point-charge force and field



This package consolidates the saved Canvas pages and worksheet scans into a conceptual progression. Each section gives the main ideas, a hallmark equation, the physical reasoning behind its derivation, misconceptions, and an immediate conceptual check with answer. Textbook supplements are labeled in the source map.

Contents and coverage map

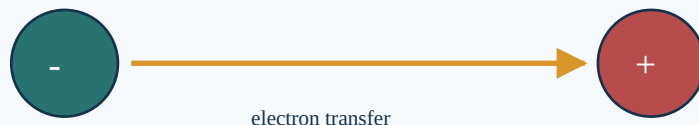
01. Charge and conservation
02. Coulomb force and electric fields
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04. Continuous charge distributions and integration
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06. Applications of Gauss law
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The companion YAML records source pages, equations in normalized LaTeX, assumptions, limiting cases, diagrams, checks, and question-generation tags. The page-list-only gaps identified during inventory are covered as supplemental sections where a saved capture was absent.

01. Charge and conservation

Scope and source status: Charge types, quantization, conservation, friction, contact, induction, conductors/insulators, electrostatics versus electrodynamics.
Question-generation tags: fill-in explain-phenomenon qualitative-prediction

Electron transfer and conservation



Core understanding

- Charge is a signed, quantized property carried mainly by protons and electrons. A neutral object has equal positive and negative charge, not an absence of charge.
- Charging by friction or contact transfers electrons; charging by induction separates charge inside an object and requires grounding to leave a net charge.
- Conservation applies to the complete isolated system: one object gaining electrons means another loses the same electrons. Electrostatics describes settled charges; electrodynamics describes charge in motion or changing fields.

Hallmark equation

$$Q_{\text{total}} = \text{constant}$$

Symbols: Q_{total} is the algebraic sum of all positive and negative charge in the chosen isolated system. **Physical meaning:** the equation says electrons may move between objects, but the system cannot gain or lose net charge. A local charge change is balanced by an equal and opposite change elsewhere.

Concept check: An initially neutral object gains electrons from a second object. Describe the final signs of both objects and explain how the total charge of the complete system changes.

Answer and explanation: An isolated object that gains electrons becomes more negative. The charge of the object changes, but the charge of the complete isolated system does not: electrons have moved from one part of the system to another. Induction is therefore a redistribution of existing charge, not creation of new charge.

02. Coulomb force and electric fields

Scope and source status: Inverse-square force, source/test charge roles, field direction, units, gravity analogy, and force from a field.

Question-generation tags: fill-in explain-phenomenon derivation

Point-charge force and field



Core understanding

- Coulomb law gives the force between point charges. The force acts along the line joining the charges, attracts unlike signs, repels like signs, and falls as $1/r^2$.
- Electric field is force per unit positive test charge. It belongs to the source arrangement, while the actual force on a charge is $F=qE$ and reverses direction for $q<0$.
- The gravity analogy is useful for the inverse-square geometry, but electric charge has two signs and can produce either attraction or repulsion.

Hallmark equation

$$F = q_1 q_2 / (4\pi\epsilon_0 r^2) \quad E = F/q_{\text{test}}$$

Symbols: q_1, q_2 are Source charges, r is their separation, ϵ_0 is the vacuum permittivity, and $\mathbf{r}_{\text{hat}}$ gives direction. E is force per unit positive test charge. **Physical meaning:** inverse-square spreading weakens interaction with distance; the signs determine attraction or repulsion, while E describes the source before a test charge is added.

Concept check: A negative test charge is placed at a point where the electric field points east. Predict the force direction and explain why field direction is defined with a positive test charge.

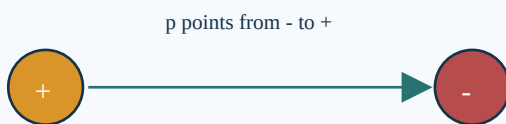
Answer and explanation: The force is opposite the field because $F=qE$ and q is negative. The field direction is defined using a positive test charge, so a negative charge reverses that direction. This is why field and force must not be treated as interchangeable vectors.

03. Field lines, superposition, and vectors

Scope and source status: Vector addition, components, symmetry cancellation, field-line density and direction, point-charge limits, dipole far-field behavior.

Question-generation tags: diagram-matching draw-phenomenon graph-interpretation

Vector superposition and dipole



Core understanding

- Superposition means calculate each source field at the observation point and add the vectors. Components cancel or reinforce according to geometry; magnitudes alone are not enough.
- Field lines are a drawing convention: the tangent gives the direction for a positive test charge, and line density represents relative strength. Lines never cross.
- Dipole fields show cancellation clearly: far away, the fields of $+q$ and $-q$ nearly cancel, so the remaining field decreases faster with distance than a single-charge field.

Hallmark equation

$$\mathbf{E}_{\text{net}} = \sum_i \mathbf{E}_i$$

Symbols: $\mathbf{E}_i$ is the field from source i and $\mathbf{E}_{\text{net}}$ is the vector total. **Physical meaning:** each source acts independently, then the components combine. This equation explains cancellation on symmetry axes and reinforcement between opposite charges.

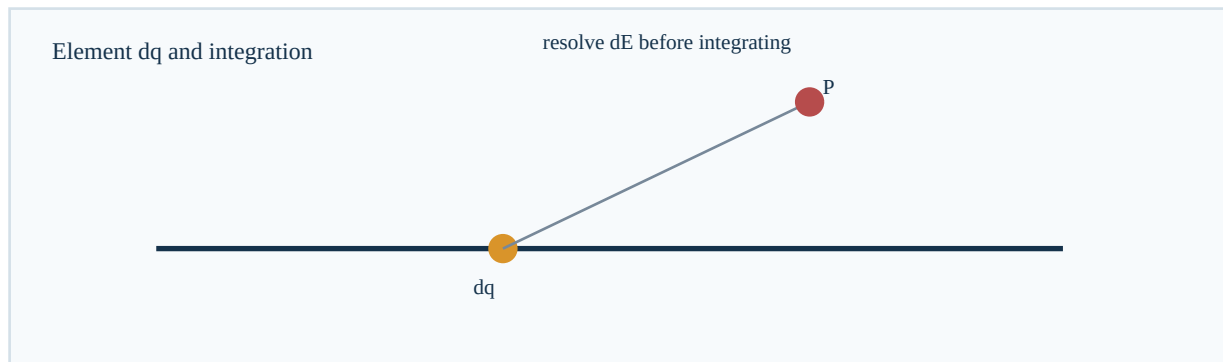
Concept check: Two equal positive charges are placed symmetrically about a point. Explain which field components cancel at the midpoint, and describe what changes if the charges have opposite signs.

Answer and explanation: Nonzero contributions can sum to zero when equal vectors point in opposite directions. Field-line density represents relative magnitude, while the tangent gives the direction a positive test charge would move. Dipole cancellation is why its distant field falls faster than a single-charge field.

04. Continuous charge distributions and integration

Scope and source status: lambda, sigma, rho, dq elements, total charge, line/arc/ring/disk/sheet/cylinder geometry, trigonometric substitutions, differential reasoning.

Question-generation tags: derivation draw-phenomenon fill-in



Core understanding

- A continuous charge distribution is handled by dividing the source into elements dq . Use $dq = \lambda dl$, σdA , or ρdV according to whether charge lies on a line, surface, or volume.
- For each element, write its distance and direction to the field point, resolve the vector into components, and integrate over the actual source limits.
- Symmetry is part of the solution: paired elements can cancel one component before integration. Trigonometric substitutions simplify the distance factor, but never replace the geometry or bounds.

Hallmark equation

$$dq = \lambda dl = \sigma dA = \rho dV \quad d\mathbf{E} = (1/4\pi\epsilon_0) dq/r^2 \hat{\mathbf{r}}$$

Symbols: λ , σ , and ρ are line, surface, and volume charge densities; dl , dA , and dV are source elements; dq is the charge in one element. **Physical meaning:** integration is a limit of adding many small Coulomb contributions whose distances and directions vary across an extended source.

Concept check: For a uniformly charged rod observed from its perpendicular bisector, explain how you would choose dq , which field component cancels, and why the remaining component must be integrated.

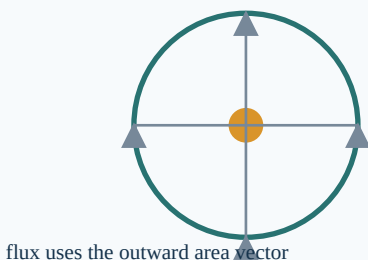
Answer and explanation: For a uniform rod, $dq = \lambda dx$. In a symmetric ring or rod arrangement, each element has a partner whose transverse component cancels. The remaining component is integrated after expressing r and the angle in terms of the chosen coordinate.

05. Electric flux and Gauss law

Scope and source status: Flux, area vectors, enclosed charge, closed surfaces, spherical/cylindrical/planar symmetry, and external-charge cancellation.

Question-generation tags: diagram-matching explain-phenomenon derivation

Closed Gaussian surface



Core understanding

- Flux is the field component normal to a surface multiplied by area, summed over the surface. The area vector points outward on a closed surface.
- Gauss law says net closed-surface flux measures enclosed charge. External charges can change the field at individual patches but add zero net flux because their field enters and leaves the surface.
- Gauss law becomes a practical field solver only when symmetry makes E constant on a chosen Gaussian surface or makes parts of the flux zero.

Hallmark equation

$$\Phi_E = \oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}} / \epsilon_0$$

Symbols: $d\mathbf{A}$ is an outward area vector, E is the local field, Q_{enc} is net enclosed charge, and ϵ_0 is vacuum permittivity.

Physical meaning: the total excess of field passing outward through any closed surface is set only by charge inside it.

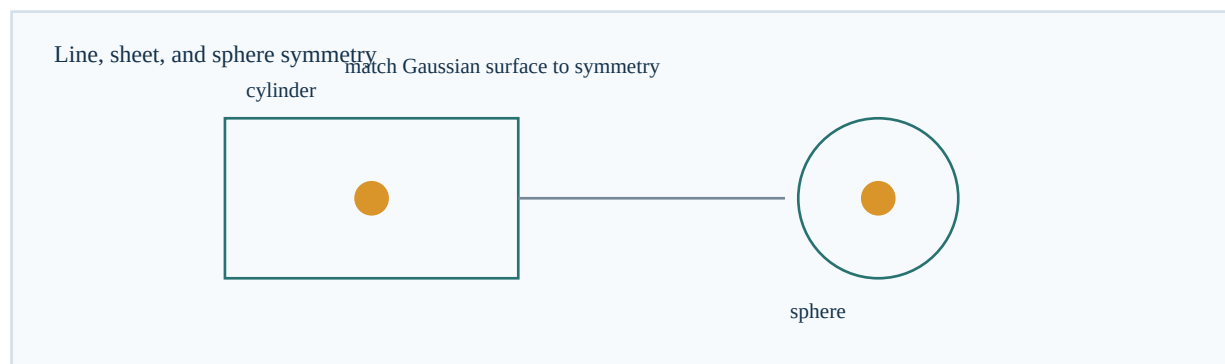
Concept check: A closed Gaussian surface encloses zero net charge but passes through a nonzero electric field. Explain how the net flux can still be zero.

Answer and explanation: The net flux is zero when the enclosed net charge is zero, even if the field is nonzero at individual points. External field lines enter and leave the closed surface, so their inward and outward contributions cancel in the total flux.

06. Applications of Gauss law

Scope and source status: Infinite line, sheet, sphere, nonuniform spherical charge, conductor surface, inside/outside field graphs.

Question-generation tags: derivation graph-interpretation qualitative-prediction



Core understanding

- For an infinite line, cylindrical area grows as r , giving E proportional to $1/r$. For a sphere, area grows as r^2 , giving the familiar inverse-square outside field.
- Inside a uniformly charged solid sphere, enclosed charge grows as r^3 while Gaussian area grows as r^2 , so E grows linearly with r .
- The same procedure works for sheets, shells, and conductors: determine enclosed charge first, then use the surface on which symmetry controls the flux.

Hallmark equation

$$E_{\text{line}} = \lambda / (2\pi\epsilon_0 r) \quad E_{\text{sheet}} = \sigma / (2\epsilon_0)$$

Symbols: λ is line charge density, σ is sheet density, r is distance from the symmetry axis or center, and ρ is volume density. **Physical meaning:** the field falloff follows how the available flux area grows: cylindrical area gives $1/r$, spherical area gives $1/r^2$, and a plane gives constant area per unit length.

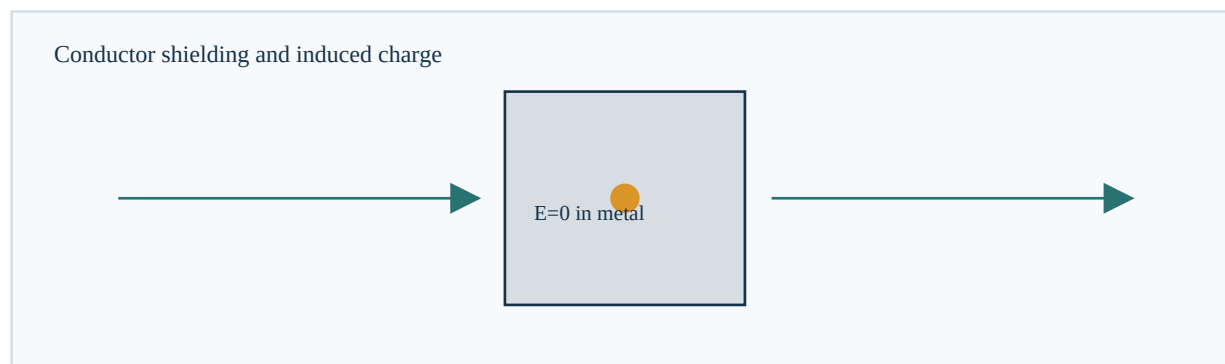
Concept check: Compare the field-distance dependence for an infinite line charge and a uniformly charged spherical shell outside the shell. Explain why their powers of r differ.

Answer and explanation: Doubling distance from an infinite line halves the field because cylindrical area grows in proportion to r . Inside a charged spherical shell, a Gaussian sphere encloses no net charge, so Gauss law gives $E=0$.

07. Conductors, shielding, and induced charge

Scope and source status: Electrostatic equilibrium, zero interior field, surface charge, cavities, shielding, pointed conductors, molecular induction.

Question-generation tags: draw-phenomenon explain-phenomenon fill-in



Core understanding

- Free charges in a conductor move until the electric field inside the metal is zero. Any remaining tangential field would continue driving surface charge, so the equilibrium field is normal to the surface.
- Excess charge resides on surfaces. A charge in a cavity induces opposite charge on the inner wall; a neutral conductor then places compensating charge on its outer surface.
- A closed conductor shields its cavity from external static fields. Sharp points have high surface-charge density and therefore unusually large nearby fields.

Hallmark equation

$$E_{\text{metal}} = 0 \quad E_{\text{outside}} = \sigma / \epsilon_0$$

Symbols: E_{metal} is the field inside conducting material and σ is surface charge density. **Physical meaning:** free charges rearrange until no force remains in the metal; the normal surface field is the field produced by the accumulated surface charge just outside.

Concept check: Explain why a static electric field cannot remain inside conducting material, and describe the induced charge on the inner wall when a positive charge is placed in a neutral cavity.

Answer and explanation: A tangential field would exert a force along the conductor and keep free charges moving, contradicting electrostatic equilibrium. A $+Q$ charge in a cavity induces $-Q$ on the cavity wall so that the conducting material itself encloses zero net charge.

08. Electric potential, energy, and equipotentials

Scope and source status: $V=U/q$, work, potential differences, point and continuous sources, equipotential surfaces, $E=-\text{grad } V$.

Question-generation tags: fill-in diagram-matching qualitative-prediction derivation

Equipotential map and field direction



E is perpendicular to equipotentials

Core understanding

- Potential V is scalar energy per charge, so point-charge potentials add algebraically. Potential differences are physical even though the zero of V is chosen by convention.
- The field does work when a charge moves through a potential difference: $\Delta U = q \Delta V$. Equipotential motion has $\Delta V = 0$ and therefore zero electric work.
- $E = -\text{grad } V$ means the field points in the direction of steepest decrease of potential. Closely spaced equipotentials indicate a stronger field.

Hallmark equation

$$\mathbf{E} = -\text{grad } V \quad \Delta U = q \Delta V$$

Symbols: V is potential energy per charge, U is potential energy, q is the charge being moved, and ∇V describes spatial change in potential. **Physical meaning:** the field points in the direction where a positive charge loses potential energy most rapidly; equipotential motion has no potential-energy change.

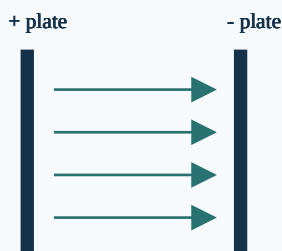
Concept check: Explain what it means physically for electric potential to decrease in a direction, and why the electric field is perpendicular to an equipotential surface.

Answer and explanation: If V decreases as x increases, dV/dx is negative and $E_x = -dV/dx$ is positive. Motion along an equipotential has $\Delta V = 0$, so $\Delta U = q \Delta V = 0$ and the electric field does no work.

09. Capacitance and dielectrics

Scope and source status: Capacitor geometry, series/parallel combinations, energy, battery-connected versus isolated cases, polarization and permittivity.
Question-generation tags: matching fill-in qualitative-prediction

Parallel-plate capacitor



Core understanding

- Capacitance is the charge stored per potential difference, determined by geometry and dielectric material. Parallel plates have $C = \epsilon A/d$ when edge fringing is negligible.
- Parallel capacitors share voltage and add capacitance; series capacitors share charge magnitude and add reciprocals. The battery-connected or isolated condition determines what changes when a dielectric is inserted.
- A dielectric polarizes: bound charges reduce the internal field for fixed free charge and allow more charge to be stored at the same voltage. Stored energy is energy in the electric field.

Hallmark equation

$$C = Q/V \quad U = \frac{1}{2} C V^2$$

Symbols: C is capacitance, Q is stored free charge, V is the potential difference, and U is stored field energy. **Physical meaning:** geometry and material determine how much charge a capacitor can hold for a given voltage; a dielectric polarizes and changes that charge-voltage relationship.

Concept check: Explain what capacitance measures, how parallel and series connections differ physically, and why inserting a dielectric changes the stored charge or voltage depending on whether a battery remains connected.

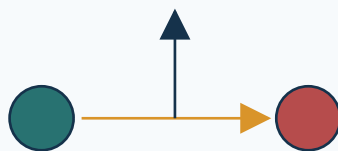
Answer and explanation: In parallel, every capacitor has the same voltage and the stored charges add, so capacitances add. A dielectric polarizes: bound charge reduces the internal field for fixed free charge and increases the capacitance.

10. Dipoles, torque, polarization, and displacement

Scope and source status: Dipole moment direction, torque, potential energy, polar/nonpolar molecules, P, D, susceptibility, free versus bound charge.

Question-generation tags: diagram-matching explain-phenomenon derivation

Dipole torque and polarization



torque rotates p toward E

Core understanding

- A dipole has equal and opposite charges separated by d , with $p=qd$ directed from negative to positive. In a uniform field, opposite forces can cancel while their separated lines of action create torque.
- Torque aligns p with E , and $U=-p \cdot E$ makes parallel alignment stable. In a nonuniform field, a dipole can also translate toward stronger field.
- Polarization P is dipole moment per volume. $D=\epsilon_0 E+P$ separates the free-charge bookkeeping from the material response; P and bound charge are not additional free charge.

Hallmark equation

$$\mathbf{p} = q \mathbf{d} \quad \tau = p E \sin\theta \quad \mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$

Symbols: p is dipole moment, d is charge separation, τ is torque, E is external field, P is dipole moment per volume, and D combines free-charge field with polarization. **Physical meaning:** separated opposite forces form a turning couple, while microscopic dipole alignment produces a macroscopic material response.

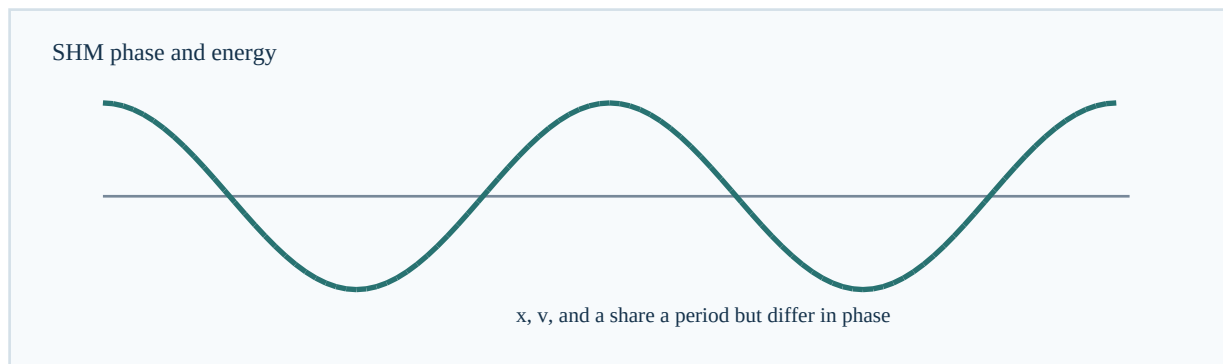
Concept check: Explain how a dipole can have zero net force but nonzero torque in a uniform electric field, and identify the stable orientation of its dipole moment.

Answer and explanation: The stable orientation is p parallel to E because $U=-pE \cos(\theta)$ is then smallest. Polarization P is the net dipole moment per unit volume, not the total free charge density.

11. Oscillations and simple harmonic motion

Scope and source status: Hooke law, SHM differential equation, amplitude, phase, period, $x/v/a$ phase, energy, phasors, resonance background.

Question-generation tags: graph-interpretation draw-phenomenon derivation



Core understanding

- SHM requires a restoring acceleration proportional to displacement: $x \ddot{=} -\omega^2 x$. A mass-spring system has $\omega = \sqrt{k/m}$, so stiffness speeds oscillation and mass slows it.
- The phase of x , v , and a matters: v is a quarter-cycle shifted from x , while a is opposite x . Turning points have zero speed and maximum restoring acceleration.
- Energy moves between kinetic and spring potential while the total remains constant in ideal SHM. Damping removes energy; driving near the natural frequency produces resonance.

Hallmark equation

$$x = A \cos(\omega t + \phi) \quad \omega = \sqrt{k/m}$$

Symbols: A is amplitude, ω is angular frequency, t is time, ϕ is initial phase, k is spring stiffness, and m is mass.

Physical meaning: the sinusoid is the motion whose acceleration always points toward equilibrium and whose restoring strength grows with displacement.

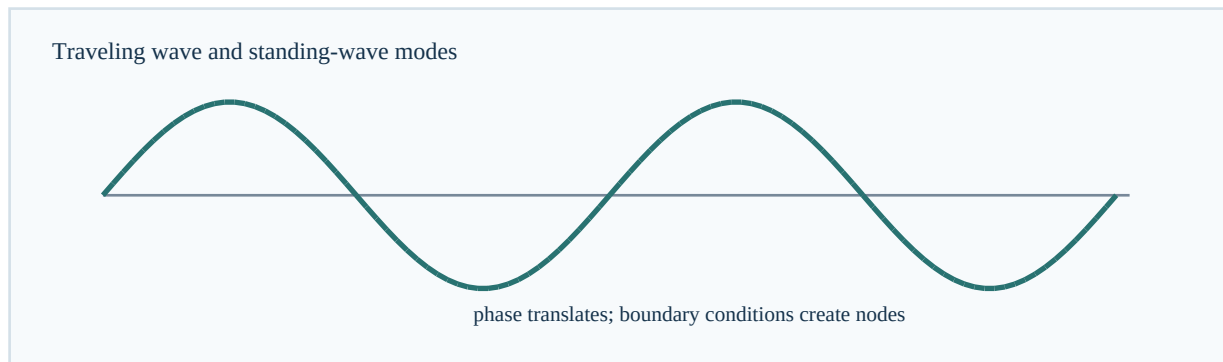
Concept check: At the positive turning point of an ideal mass-spring oscillator, describe the velocity, acceleration, spring energy, and direction of the restoring force.

Answer and explanation: At maximum positive displacement, velocity is zero and acceleration is maximum negative, pointing toward equilibrium. Increasing mass increases the period because $T = 2\pi \sqrt{m/k}$.

12. Mechanical waves, interactions, and standing waves

Scope and source status: Pulses, traveling wave translation, transverse/longitudinal motion, wave speed, linear wave equation, reflection, transmission, interference, standing waves.

Question-generation tags: fill-in diagram-matching derivation draw-phenomenon



Core understanding

- A traveling wave carries energy while each medium element oscillates locally. In $y=f(x-vt)$, constant phase moves right; the pattern speed is not the same as the particle speed.
- The harmonic form identifies A , k , ω , phase, wavelength, frequency, and direction. The linear wave equation relates spatial curvature to temporal acceleration when the medium has constant wave speed.
- Reflection depends on the boundary, interference adds displacements, and two opposite waves form standing nodes and antinodes. Fixed-end conditions select $\lambda_n = 2L/n$.

Hallmark equation

$$y = A \cos(kx - \omega t + \phi) \quad v = f\lambda$$

Symbols: A is displacement amplitude, k is wave number, ω is angular frequency, ϕ is phase, λ is wavelength, f is frequency, and v is waveform speed. **Physical meaning:** the phase argument tracks a repeating pattern in both space and time; the sign before ωt selects travel direction and $v=f\lambda$ links spatial and temporal repetition.

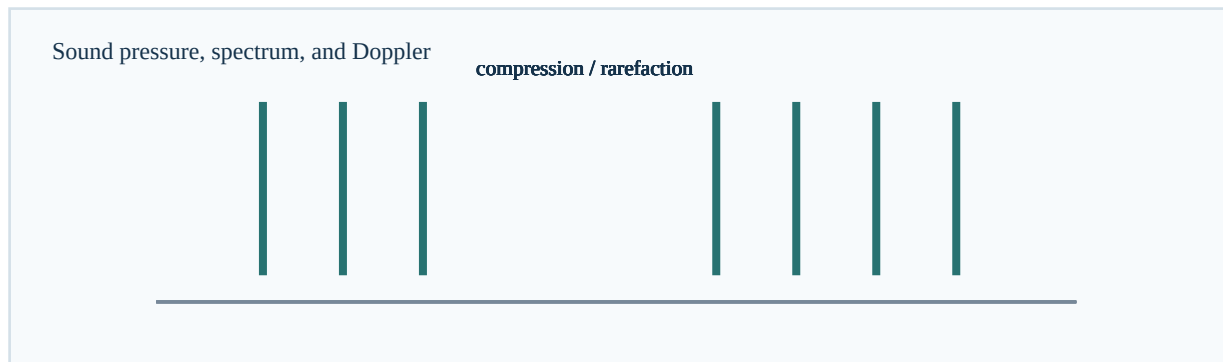
Concept check: For $y=f(x-vt)$, explain how you know the wave moves right, and distinguish the speed of the waveform from the velocity of a point in the medium.

Answer and explanation: The form $f(x-vt)$ moves right because constant phase requires x to increase with t . At a boundary the source-set frequency remains continuous, while speed and wavelength adjust to the new medium.

13. Sound, intensity, spectra, hearing, and Doppler effect

Scope and source status: Sound production/detection, compression and rarefaction, pressure amplitude, intensity, decibels, harmonics, noise, hearing, Doppler shift.

Question-generation tags: fill-in graph-interpretation explain-phenomenon



Core understanding

- Sound is a longitudinal pressure disturbance: compressions are high-pressure regions and rarefactions are low-pressure regions. Sound needs a material medium.
- Intensity is power per area and falls as $1/r^2$ for an isotropic source. Decibels describe ratios logarithmically, so equal dB changes represent multiplicative intensity changes.
- Frequency sets pitch, amplitude and pressure relate to loudness, and harmonic content sets timbre. Relative motion changes wavefront arrival rate: approach raises perceived frequency and recession lowers it.

Hallmark equation

$$I = P/(4\pi r^2) \quad \beta = 10 \log_{10}(I/I_0)$$

Symbols: I is intensity, P is acoustic power, r is distance, β is sound level, and I_0 is the reference intensity. **Physical meaning:** spherical spreading lowers intensity with distance, while the logarithm expresses how hearing compares multiplicative intensity ratios. Doppler shifts arise from arrival rate, not from a change in source power.

Concept check: An observer moves toward a stationary sound source. Predict the change in perceived frequency and explain it using wavefront arrival rate rather than source loudness.

Answer and explanation: Doubling the distance quarters intensity because the same power spreads over four times the spherical area. An approaching source or observer increases wavefront arrival rate, so perceived frequency rises.

References and provenance

Primary: saved SingleFile Canvas pages in docs/reference/canvas_pages and worksheet scans in docs/reference/reference-texts/Physics 2 scans Test 1. Supplemental: OpenStax University Physics Volume 1, Chapters 15-17; OpenStax University Physics Volume 2, Chapters 5-7; Young and Freedman, University Physics, relevant electrostatics, oscillations, waves, and sound chapters.

Original captures and scans were preserved unchanged. Repeated page titles were normalized in the knowledge base. Figures in this package are clean instructional redraws; the source inventory records where professor visuals are available for later extraction and reuse.